# Cover Letter to Reviewer 3

**Round 2 Revision**

**Manuscript:** "Content-Conditioned Invariant Feature-Statistic Augmentation: Unifying Robustness and Weight Averaging for Domain Generalization" (CIFA)
**Journal:** *Electronics* (MDPI) **Manuscript ID:** electronics-4654208
**Date:** 3 November 2026

Dear Reviewer 3,

We sincerely thank you for the detailed and constructive assessment and for recognising the substantial improvement in the mathematical formulation, experimental validation, statistical analysis, and interpretation. We have addressed the three limited revisions. Each point below is preceded by a location tag giving the exact page and line range in the tracked PDF, which uses continuous line numbering.

## Point 1 (R3-1) — Mathematical justification for the conditional posterior bound

[**Tracked PDF: Page 8, Line 304 through Page 9, Line 348; proofs on Page 10, Line 354 through Page 11, Line 400**]

Your comment is correct, and we have restructured Proposition 1 rather than defend the earlier statement. The data-processing inequality controls the distributions (laws) of the posterior random variables, not the posterior evaluated at each fixed observation; at rare observations a small likelihood perturbation can be amplified. The proposition is now stated at two clearly separated levels:

- **(L2-i) Exact case.** When $\tau = 0$ and the partner source is exogenous, the content posterior is preserved exactly;
- **(L2-ii)(a) Distribution-level bounds (unconditional).** The coupling gives the only unconditional statements:

$$\mathrm{TV}(P_{C,Z}, P_{C,T(Z)}) \le \tau, \qquad \mathrm{TV}(p(Z), p(T(Z))) \le \tau;$$

- **(L2-ii)(b) Pointwise bound (conditional).** Retained only under an explicit inter-model likelihood-ratio condition $|\log p(\mathbf{b}, r, \mathbf{d} \mid c) - \log p(\mathbf{b}, r, \mathbf{d}_{aug} \mid c)| \le \alpha$, under which $\mathrm{TV}(p(C \mid Z), p(C \mid T(Z))) \le \tanh \alpha$.

The KL corollary is now conditional on the same assumption, $D_{\mathrm{KL}} \le \tanh(\alpha)/q_{\min}$, and the earlier $\tau \log(M/p_{\min})$ and $\tau/q_{\min}$ pointwise bounds are removed. The label-posterior drift $\delta$ is treated as a directly measured quantity rather than added to $\tau$. The full derivations are given in the revised proof (Pages 10–11). Theorems 1 and 2 are unaffected because they use the distribution-level total-variation quantity.

## Point 2 (R3-2) — Consistent distinction of equivalence, non-significance, and superiority

[**Tracked PDF: Page 1, Lines 17–19; Pages 21–23, Lines 795–828; Page 36, Lines 1168–1200**]

We have applied one consistent terminology throughout the twelve affected passages. Non-significant contrasts are reported as statistically tied or on par; the term equivalence is used only for the TOST procedure and always with the equivalence margin stated; and no superiority is claimed. We also corrected the TOST implementation to the standard two-one-sided procedure using 90% confidence intervals and recomputed every contrast. On PACS ($n = 10$), at the one-point margin equivalence is confirmed only against RSC; at the two-point margin it additionally holds against SWAD, CORAL, and Fish; the CIFA–EFDM contrast is neither superior nor certified equivalent. These changes appear at the abstract (Page 1), introduction and contributions (Pages 2–3), experiments and captions (Pages 21–25), appendix (Pages 33–35), and conclusions (Page 36). The conditional nature of the unification is kept explicit throughout.

### Point 3 (R3-3) — Clean manuscript and consistency check

**[Tracked PDF: Whole manuscript; hyphenation at Page 36, Line 1168]**

A clean, fully accepted version (`CIFA_Electronics_clean.pdf`, 39 pages) is provided alongside the tracked version. We have verified that every cross-reference resolves with no dangling or duplicate labels; all 27 bibliography entries are cited; equations, tables, and figures use automatic numbering with no hard-coded numbers; and the terminology and notation are consistent. The clean manuscript renders entirely in black with no revision marks. As part of this pass we also standardised the hyphenation (e.g., "content–style"). Both the clean and tracked manuscripts compile with zero errors and zero overfull boxes, and their accepted content is identical.

We thank you once again for the constructive and supportive review, and hope these revisions fully address the remaining points.

Sincerely,
Qingsong Cai (on behalf of all authors)
*Beijing Technology and Business University*
*caiqs@btbu.edu.cn*